

Encrypt It, Send It



E Remember that a message locked by one key in the pair can only be unlocked by the other key of the pair. If Amar wants to prove to everyone that a message is sent by him, he can “sign” the message by locking it using his ‘Private Key’. Since it has been locked by Amar’s ‘Private Key’, it can now only be unlocked by his ‘Public Key’.



G Anyone can unlock the message Amar sent to Birbal, using Amar’s ‘Public Key’, so his message is not secret. But because the message can be unlocked only by Amar’s ‘Public Key’, it proves to everyone that it has been locked by Amar’s ‘Private Key’. Since only Amar has his ‘Private Key’, he must have sent the message!

Quick Q&A

Q. How would Amar send Birbal a secret, signed message?

A. He would first lock it using his ‘Private Key’, then lock it using Birbal’s ‘Public Key’!



S Amar receives the message from Birbal and sees that it is locked. He pulls out his secret ‘Private Key’ and tries to unlock (decrypt) the message. Since it has been locked using his own ‘Public Key’, he can unlock it and read the secret message.

